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Table of Contents

Abstract	1
I. Introduction	2
II. Results	4
Task 1 – PCA/LDA for Quality Data (XMEAS(37-41)) Monitoring	4
Task 2 – PCA/LDA for Process Data Monitoring	19
Task 3 – CCA Using Quality Data (XMEAS(37-41))	36
Task 4 – Concurrent CCA Monitoring Using Quality Data (XMEAS(37-41))	41
iii. Conclusion & Discussion	44
Reference	45

Abstract

CHE599 provides a thoroughly study of principles of process data analytics and machine learning tools for process data, quality data, and other data acquired from sensors. This report is mainly focuses on product quality measurements through the TEP process. In this report, PLS, LDA, CCA and CCCA are performed on TEP. Meanwhile, each group is assigned with normal data and disturbance data. The analysis results through different models will be displayed in this report.

I. Introduction

Vast amounts of data are generated in process facilities from sensors that record information from all aspects of the process including flows, temperatures, and levels. On the other hand, Product quality measurements are necessary to make sure that the process produces on-spec products and operates efficiently in material and energy consumptions. The Tennessee Eastman process was established as a challenging problem for researchers to test new ideas for process monitoring, optimization, and control.

As shown in Figure 1, there are five unit operations in the TEP process: a reactor, a condenser, a flash separator, a reboiler stripper, and a recycle compressor. The process contains 12 manipulated variables and 41 measured variables. In TEP, process measurements are sampled with an interval of 3 minutes, while 19 composition measurements are sampled with time delays that vary from 6 minutes to 15 minutes. For all these monitoring schemes, XMEAS(1-22) and MV(1-11) are regarded as the process variables, where XMEAS(1-22) are the process measurements and XMV(1-11) are the manipulated variables. XMEAS(37-41) are selected as the quality variables, which are quality measurements.

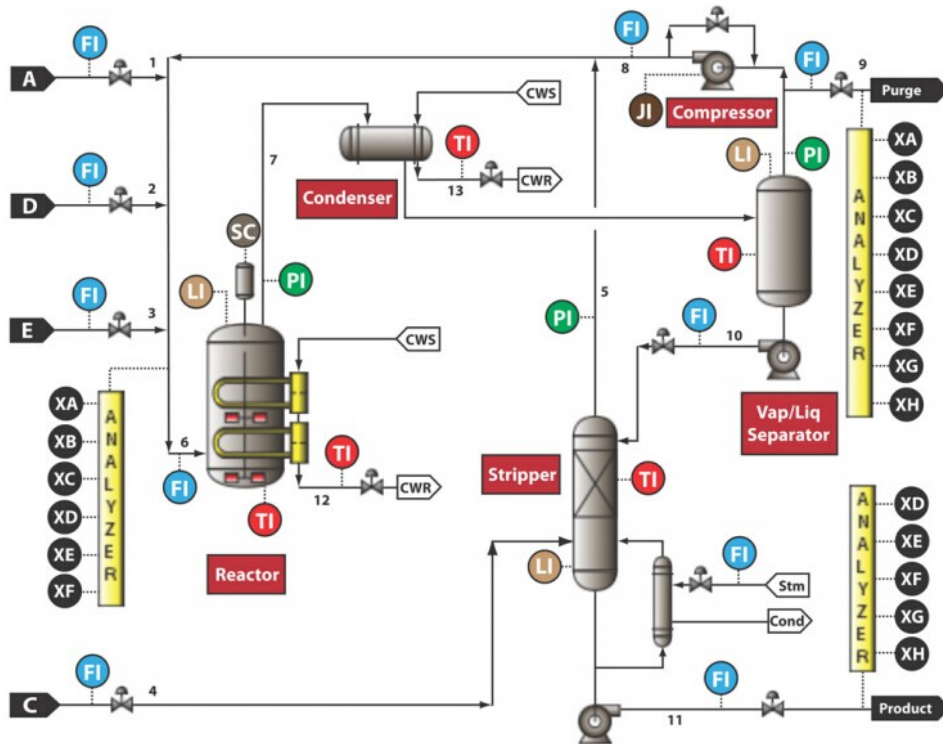


Fig.1: Schematic diagram of the Tennessee Eastman process

There are 15 known disturbances in TEP, which are listed in Fig. 2. In Fig. 2, the disturbances are divided into two groups, quality-relevant and quality-irrelevant, using selected quality variables XMEAS(37-41). Zhu et al. (2016) further give the fault detection rates for the quality-relevant disturbances and the false alarm rates for the quality-irrelevant cases.

Disturbances	Detailed Description	Type	Classification
IDV(1)	A/C feed ratio, B composition constant (stream 4)	Step	Quality-Relevant
IDV(2)	B composition, A/C ratio constant (stream 4)	Step	Quality-Relevant
IDV(3)	D feed temperature (stream 2)	Step	Quality-Irrelevant
IDV(4)	Reactor cooling water inlet temperature	Step	Quality-Irrelevant
IDV(5)	Condenser cooling water inlet temperature	Step	Quality-Relevant
IDV(6)	A feed loss (stream 1)	Step	Quality-Relevant
IDV(7)	C header pressure loss - reduced availability (stream 4)	Step	Quality-Relevant
IDV(8)	A, B, C feed composition (stream 4)	Random variation	Quality-Relevant
IDV(9)	D feed temperature (stream 2)	Random variation	Quality-Irrelevant
IDV(10)	C feed temperature (stream 4)	Random variation	Quality-Relevant
IDV(11)	Reactor cooling water inlet temperature	Random variation	Quality-Irrelevant
IDV(12)	Condenser cooling water inlet temperature	Random variation	Quality-Relevant
IDV(13)	Reaction kinetics	Slow drift	Quality-Relevant
IDV(14)	Reactor cooling water valve	Sticking	Quality-Irrelevant
IDV(15)	Condensor cooling water valve	Sticking	Quality-Irrelevant

Fig.2: TEP disturbances or fault cases.

Our group was assigned with data for the normal case where no disturbances are introduced, with 5 cases[**IDV(3)**, **IDV(5)**, **IDV(9)**, **IDV(13)** and **IDV(14)**] where disturbances are introduced.

II. Results

Task 1 – PCA/LDA for Quality Data (XMEAS(37-41)) Monitoring

1. Build a PCA model based on the normal case data, and establish T^2 and Q control limits Quality Data (XMEAS(37-41)).

- 1) First transform d00.dat to excel file and transpose it to form a 52×500 matrix with 52 variables and 500 observations. Then we delete columns XMEAS(23~36), and rearrange the left columns in the order of XMEAS(1~22), XMV(1~11) and XMEAS(37~41), which gives us a 38×500 matrix, and this file is denoted as "d00Transpose". We perform the similar rearrangement to the five test data sets. Use R-studio to load "d00Transpose".
- 2) Perform PCA on the Quality Data (XMEAS(37-41)).

```
> pve
```

```
[1] 0.2436209 0.2167004 0.2084635 0.1785856 0.1526296
```

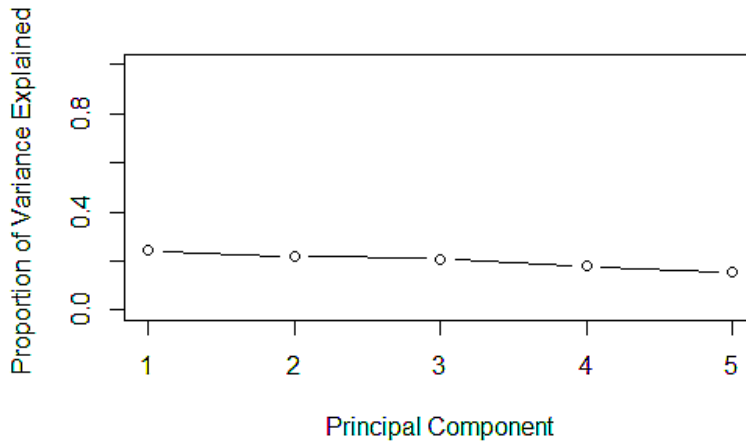


Fig. 3: PVE of training set of Quality Data

From the value of pve and the plot, we can see that all principle components must be used in order to explain 95% variance of the original dataset. However, it's meaningless to select all PCs to obtain a full model, therefore, the number of PCs is selected as 4, which can explain 84.74% variance.

- 3) Calculate the control limits for T^2 and SPE based on the PCA model.

The control limit for SPE is $\delta_\alpha^2 = g\chi_\alpha^2(h)$, where $g = \frac{\theta_2}{\theta_1}$ and $h = \frac{\theta_1^2}{\theta_2}$.

$$\theta_1 = \sum_{i=l+1}^m \lambda_i$$
$$\theta_2 = \sum_{i=l+1}^m \lambda_i^2$$

where:

- a) $m = 5$ = number of variables;
- b) $l = 4$ = number of PCs we selected.

The pr.var we get in the previous part is the λ_i . Therefore, the g and h can be calculated. Thus, the control limit for SPE is 2.931602 and the control limit for T^2 is 9.487729.

2. Use the fault detection control limits to perform fault detection on the cases with potential disturbances, and plot the results.

1) Monitor IDV(3):

- a) First transform d03_te.dat to excel file noted by "d03teOriginal" and load it with R-studio.
- b) Scale the test data with the mean and standard deviation of the train data.
- c) Perform PCA to the scaled test data
Again, the number of PCs is selected as 4, which can explain 87.88% variance.
- d) SPE statistic is calculated by $SPE = x^T (I - PP^T)x$.

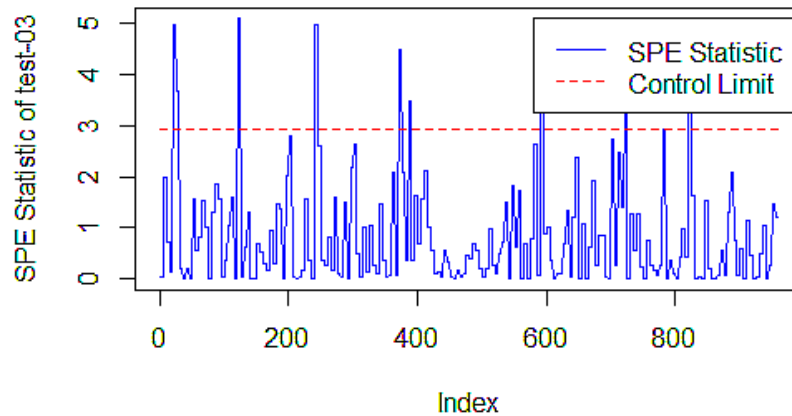


Fig. 4: SPE statistics of testing set of Quality Data IDV(3)

```
> summary(Q<SPElimit)
```

Mode	FALSE	TRUE	NA's
logical	45	915	0

From the value and plot above, we can see most samples are within the control limit of SPE.

- e) T^2 statistic is calculated by $T^2 = x^T P \Lambda^{-1} P^T x$

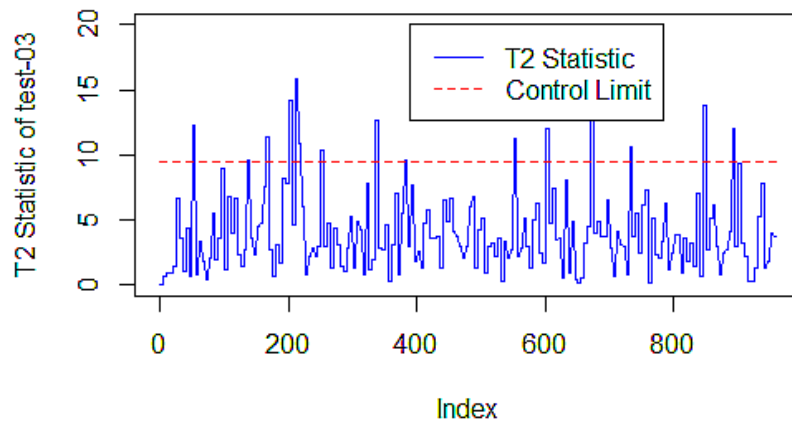


Fig. 5: T^2 statistics of testing set of Quality Data IDV(3)

```
> summary(T2<T2limit)
```

Mode	FALSE	TRUE	NA's
logical	75	885	0

From the value and plot above, we can see most samples are within the control limit of T^2 .

2) Monitor to IDV(5):

```
> pve
```

```
[1] 8.913416e-01 1.038137e-01 4.258367e-03
```

```
[4] 5.642359e-04 2.217384e-05
```

The first two PCs already explain more than 99% variance, thus the number of PCs we select is 2.

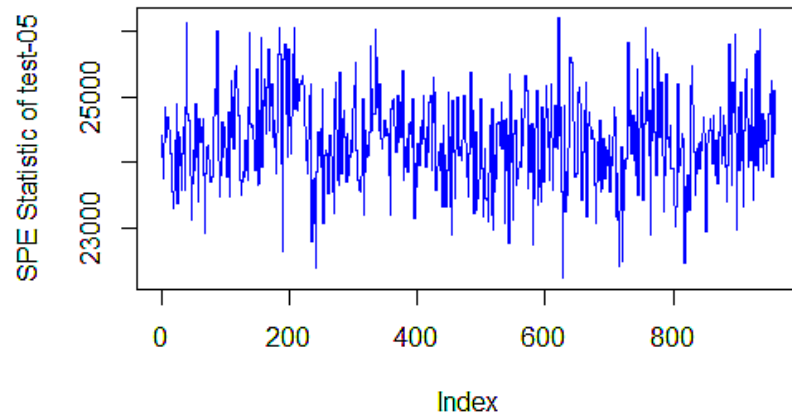


Fig. 6: SPE statistics of testing set of Quality Data IDV(5)

```
> summary(Q<SPElimit)
  Mode  FALSE   TRUE  NA's
logical  960    0    0
```

Recall that our SPElimit from normal data is 2.931602. From the value and plot above, we can see all samples are beyond the control limit of SPE.

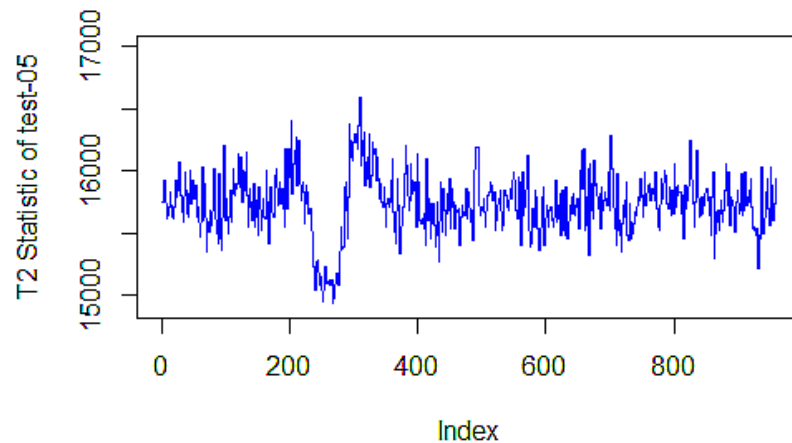


Fig. 7: T^2 statistics of testing set of Quality Data IDV(5)

```
> summary(T2<T2limit)
  Mode  FALSE   TRUE  NA's
logical  960    0    0
```

Recall that our T2limit from normal data is 9.487729. From the value and plot above, we can see all samples are beyond the control limit of T^2 .

3) Monitor to IDV(9):

```
> pve
[1] 0.3470635 0.2177488 0.1805713 0.1401411 0.1144753
```

The number of PCs is selected as 4, which can explain 88.55% variance.

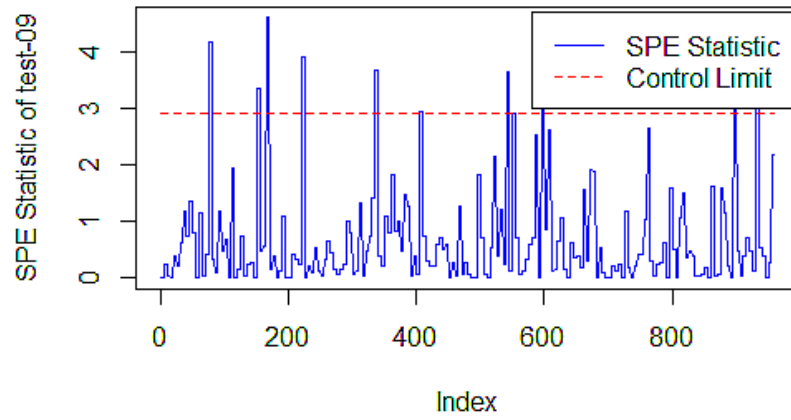


Fig. 8: SPE statistics of testing set of Quality Data IDV(9)

```
> summary(Q<SPElimit)
  Mode  FALSE   TRUE  NA's
logical    50   910     0
```

From the value and plot above, we can see only a small portion of samples are beyond the control limit of SPE.

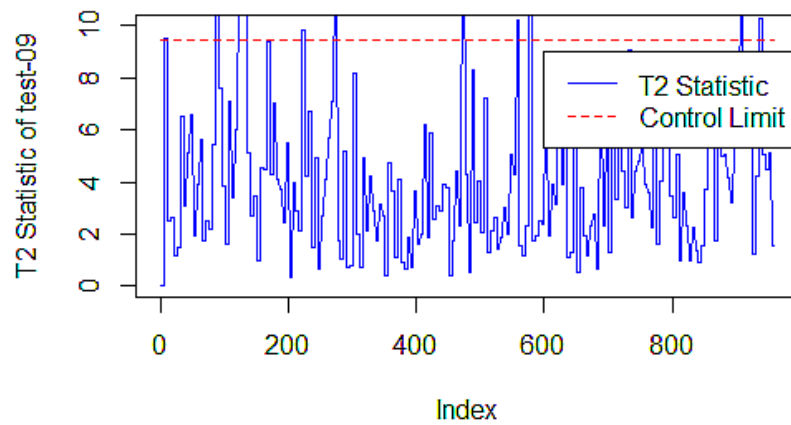


Fig. 9: T^2 statistics of testing set of Quality Data IDV(9)

```
> summary(T2<T2limit)
  Mode  FALSE   TRUE  NA's
logical    60   900     0
```

From the value and plot above, we can see that only a small portion of samples are beyond the control limit of T^2 .

4) Monitor to IDV(13):

```
> pve
```

```
[1] 0.956647358 0.018545002 0.010415071 0.008017381 0.006375188
```

The number of PCs is selected as 1, which can explain 95.66% variance.

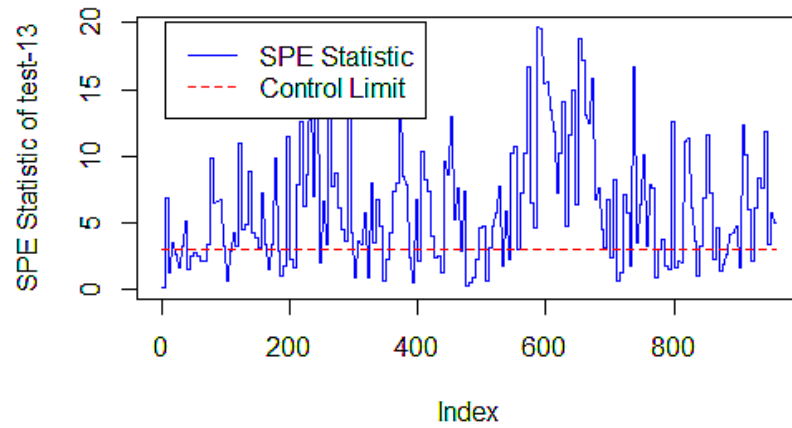


Fig. 10: SPE statistics of testing set of Quality Data IDV(13)

```
> summary(Q<SPElimit)
```

Mode	FALSE	TRUE	NA's
logical	690	270	0

From the value and plot above, we can see more than 71% samples are beyond the control limit of SPE.

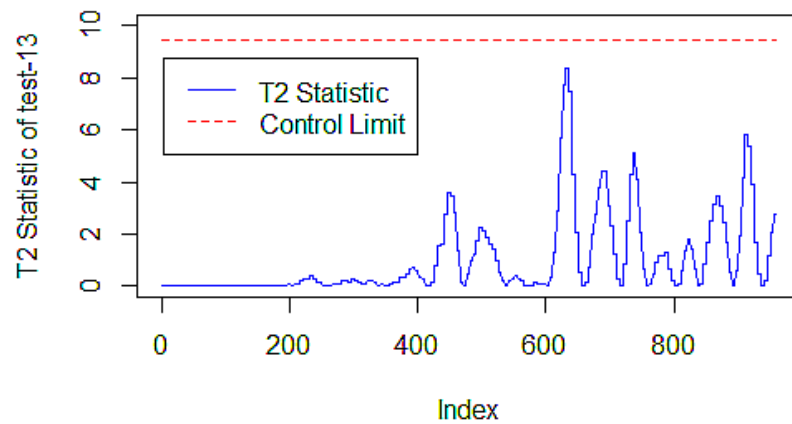


Fig. 11: T^2 statistics of testing set of Quality Data IDV(13)

```
> summary(T2<T2limit)
```

Mode	FALSE	TRUE	NA's
logical	0	960	0

From the value and plot above, we can see that all samples are within the control limit of T^2 .

5) Monitor to IDV(14):

```
> pve
```

```
[1] 0.2431368 0.2289530 0.1919683 0.1883938 0.1475481
```

The number of PCs is selected as 4, which can explain more than 85.52% variance.

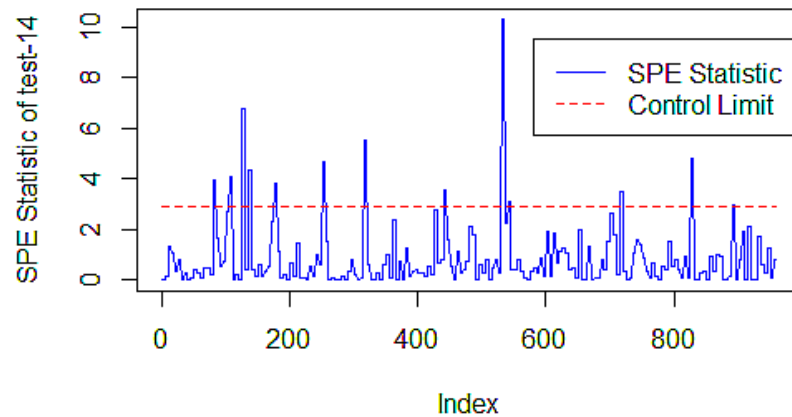


Fig. 12: SPE statistics of testing set of Quality Data IDV(14)

```
> summary(Q<SPElimit)
```

Mode	FALSE	TRUE	NA's
logical	65	895	0

From the value and plot above, we can see most samples are within the control limit of SPE.

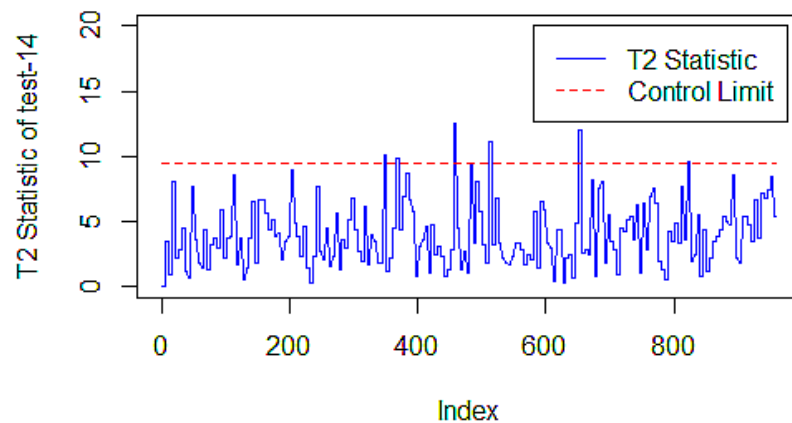


Fig. 13: T^2 statistics of testing set of Quality Data IDV(14)

```
> summary(T2<T2limit)
  Mode  FALSE   TRUE   NA's
logical  30    930     0
```

From the value and plot above, we can see that most samples are within the control limit of T^2 .

3. Compare your analysis with Fig. 2 and see if your conclusions on the quality-relevant and quality irrelevant classification agree with Fig. 2.

We first make a summary as below:

Results of quality monitoring	% of Samples Beyond SPElimit	% of Samples Beyond T2limit	Classification
IDV (3)	4.69	7.81	Quality-Irrelevant
IDV (5)	100	100	Quality-Relevant
IDV (9)	5.21	6.25	Quality-Irrelevant
IDV (13)	71.88	0	Quality-relevant
IDV (14)	6.77	3.13	Quality-Irrelevant

Table 1

Our conjecture:

- 1) As for IDV(3), IDV(9) and IDV(14), there is only a very small portion of samples beyond the corresponding control limit. Therefore, IDV(3), IDV(9) and IDV(14) fall into the classification of quality-irrelevance.;
- 2) As for IDV(5), all samples are beyond the corresponding control limit, therefore it is obviously quality-relevant.;
- 3) As for IDV(13), although there is no fault in the T^2 subspace, most samples are out of SPE control limit. Therefore, IDV(13) falls into the classification of quality-relevance.

Therefore, the above conclusions on the classification agree with Fig. 2.

4. Use the LDA classification method between the normal case and some of the cases with disturbances and see how overlapped they are, and plot the results.

```
Code sample(for IDVx):
LDAX = LDA_IDVx_1
ldax = LDAX[,37:42]

library(MASS)
lda.fit = lda(Label ~., data = ldax)
plda = predict(object=lda.fit,newdata=ldax)
plda$x
plda$class
plot(plda$x,col=plda$class)
plot(plda$x,col=ldax$Label+1)
```

1) IDV(3):

Result:

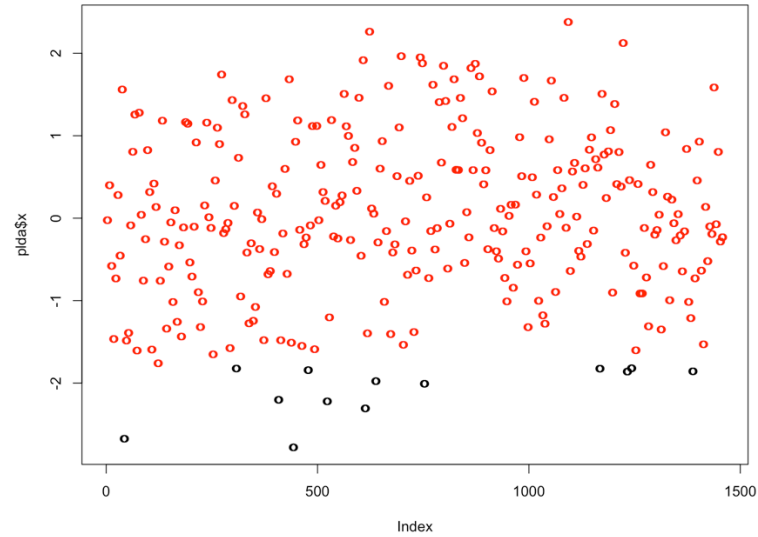


Fig. 14: LDA result for IDV(3)

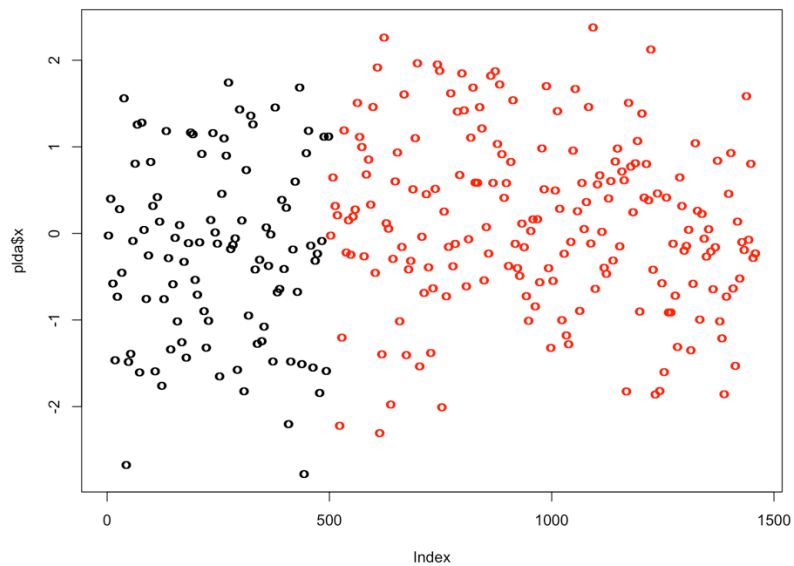


Fig. 15: Actual result for IDV(3)

From the plotting above, we can conclude that the data from IDV3 are highly overlapped with the normal data. As a result, we can not separate these two groups of data by LDA.

2) IDV(5):

Result:

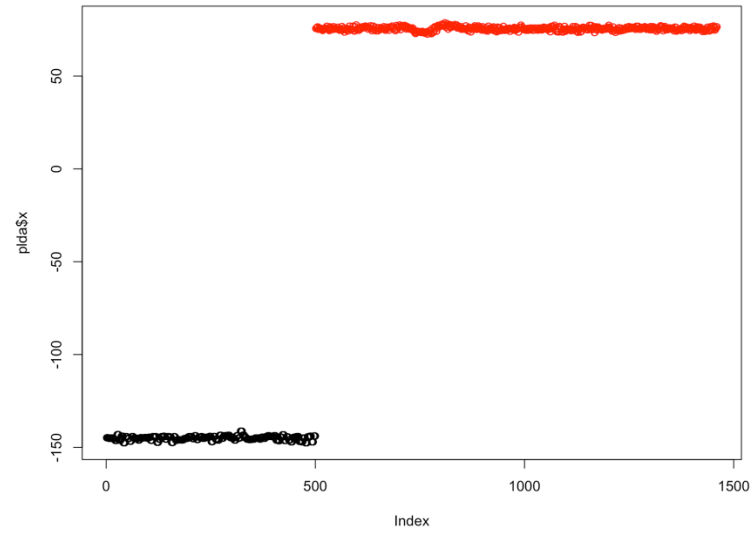


Fig. 16: LDA result for IDV(5)

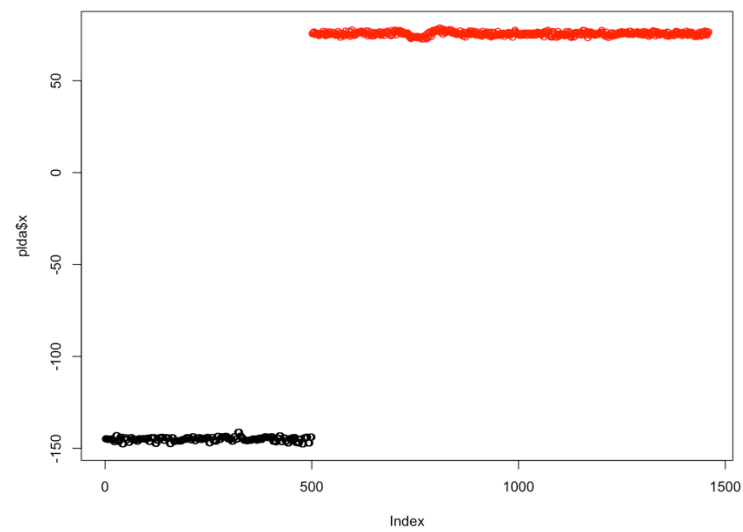


Fig. 17: Actual result for IDV(5)

From the plotting above, we can conclude that the data from IDV5 are not overlapped with the normal data. As a result, we can separate these two groups of data by LDA.

- 3) IDV(9):
Result:

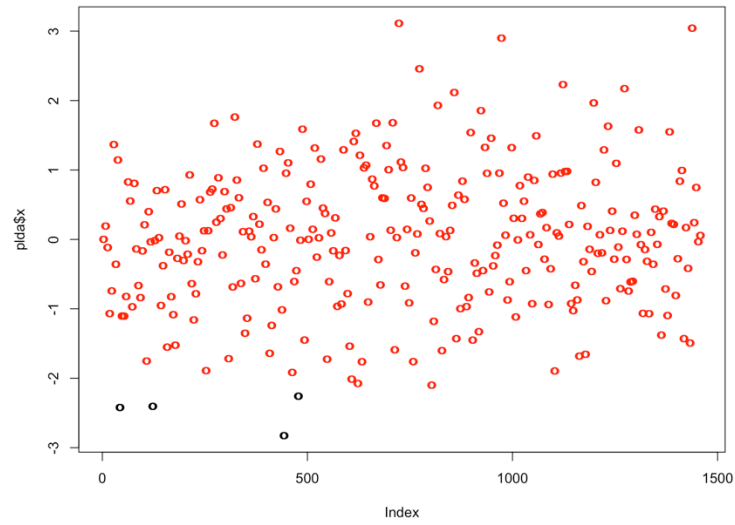


Fig. 18: LDA result for IDV(9)

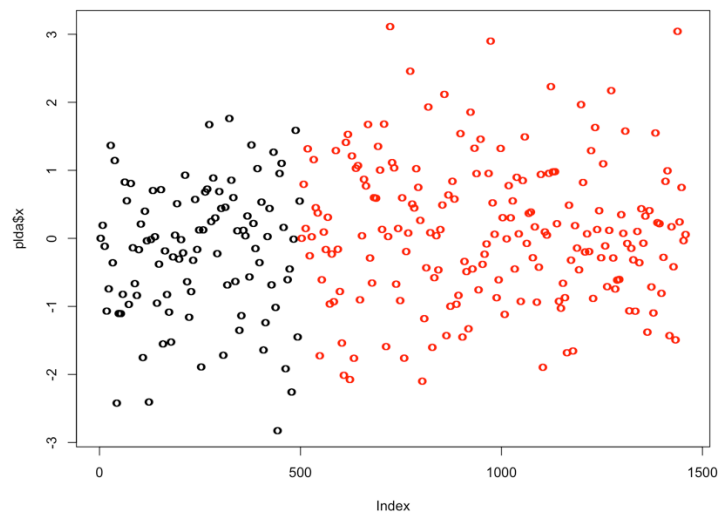


Fig. 19: Actual result for IDV(9)

From the plotting above, we can conclude that the data from IDV9 are highly overlapped with the normal data. As a result, we can not separate these two groups of data by LDA.

- 4) IDV(13):
Result:

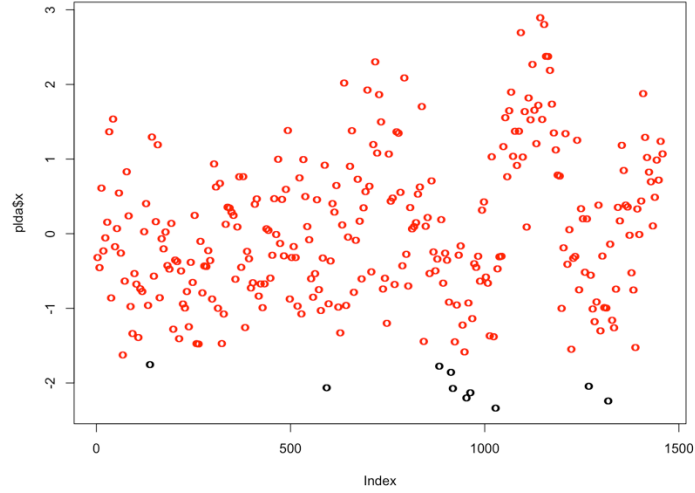


Fig. 20: LDA result for IDV(13)

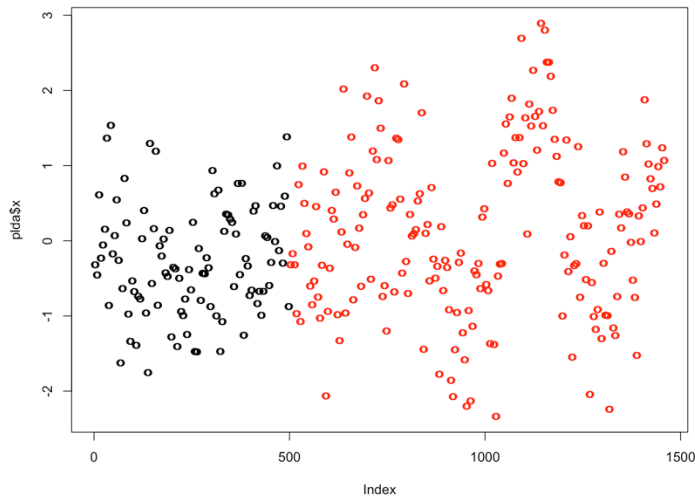


Fig.21: Actual result for IDV(13)

From the plotting above, we can conclude that the data from IDV13 are highly overlapped with the normal data. As a result, we can not separate these two groups of data by LDA.

5) IDV(14):

Result:

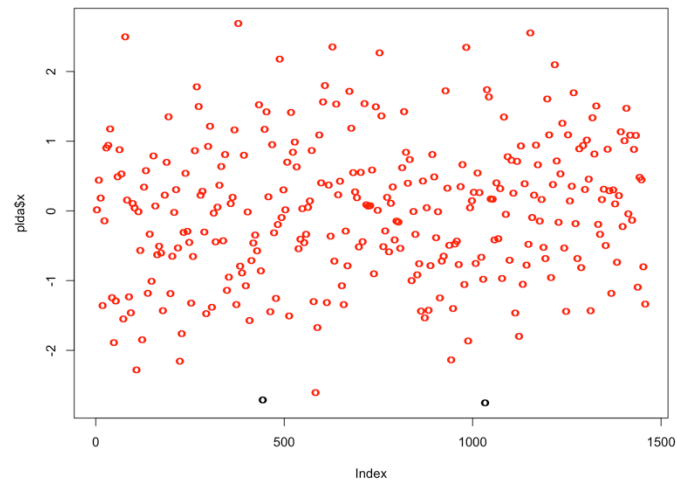


Fig. 22: LDA result for IDV(14)

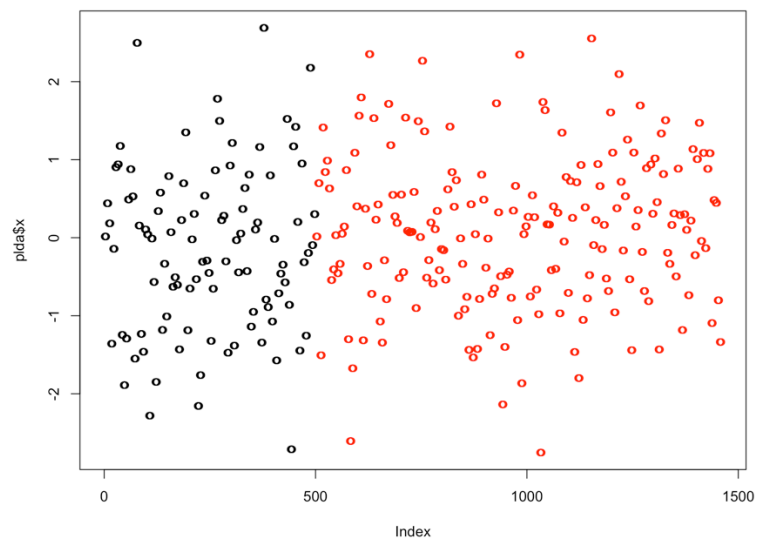


Fig. 23: Actual result for IDV(14)

From the plotting above, we can conclude that the data from IDV14 are highly overlapped with the normal data. As a result, we can not separate these two groups of data by LDA.

6) Mixed:

Result:

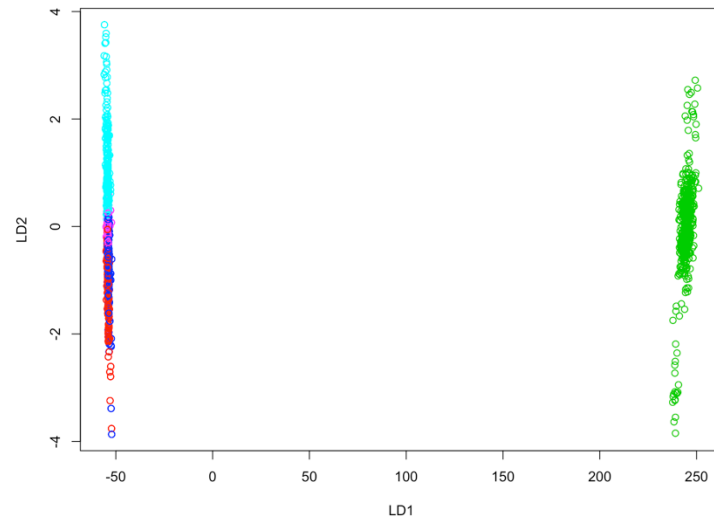


Fig. 24: LDA result for all groups(LD1&LD2)

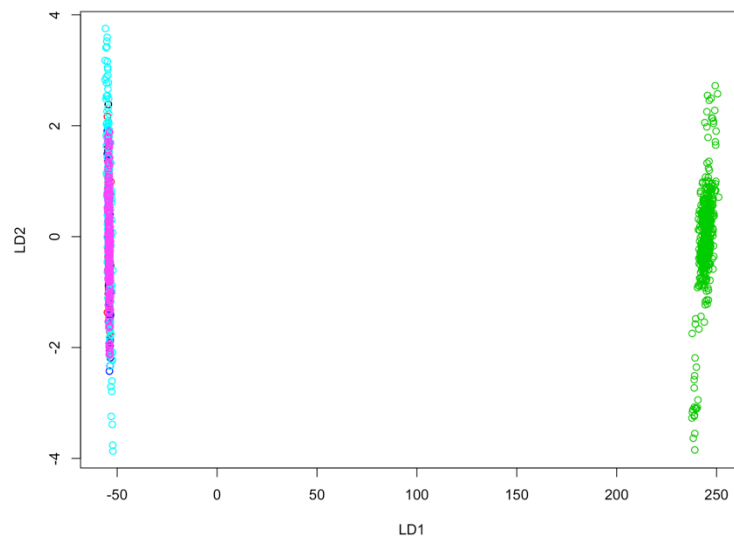


Fig. 25: Actual result for all groups(LD1&LD2)

The first two LDs can describe 99% variance.

We can also use LD2 & LD3 to plot.

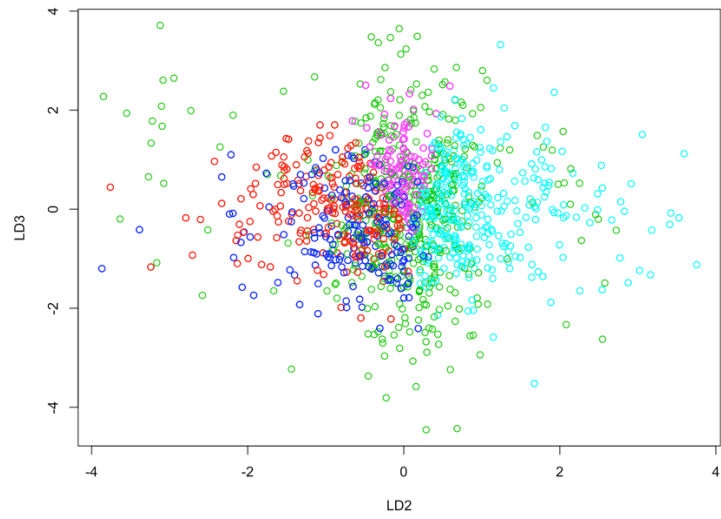


Fig. 26: LDA result for all groups(LD2&LD3)

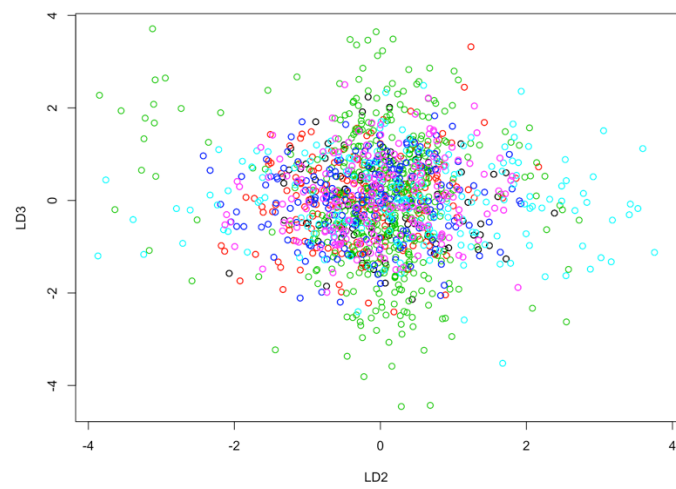


Fig. 27: Actual result for all groups(LD2&LD3)

From the plotting above, we can conclude that only one group of data (IDV5, green) are separated by LDA.

Task 2 – PCA/LDA for Process Data Monitoring

1. Build a PCA model based on the normal process data, and establish T² and Q control limits.

- 1) First delete columns of XMEAS(23~41) from normal and disturbed data. The left 33 variables are process variables: XMEAS(1~22) and XMV(1~11).
- 2) Use R-studio to load the normal data with process variables only, which is a 500×33 matrix.
- 3) Perform PCA on the Process Data, XMEAS(1~22) and XMV(1~11)

```
> cumsum(pve)
[1] 0.1638885 0.2599930 0.3392367 0.4056219
[5] 0.4676307 0.5284061 0.5850700 0.6315229
[9] 0.6766777 0.7143514 0.7473451 0.7795546
[13] 0.8098466 0.8380437 0.8648666 0.8898681
[17] 0.9135770 0.9329497 0.9512576 0.9648825
[21] 0.9771085 0.9870203 0.9942326 0.9969128
[25] 0.9983525 0.9991128 0.9995143 0.9997903
[29] 0.9999079 0.9999973 1.0000000 1.0000000
[33] 1.0000000
```

From the value of pve, the number of PCs is selected as 19, which can explain more than 95% variance.

- 4) Calculate the control limits for T² and SPE based on the PCA model.

The control limit for SPE is $\delta_{\alpha}^2 = g\chi_{\alpha}^2(h)$, where $g = \frac{\theta_2}{\theta_1}$ and $h = \frac{\theta_1^2}{\theta_2}$.

$$\theta_1 = \sum_{i=l+1}^m \lambda_i$$
$$\theta_2 = \sum_{i=l+1}^m \lambda_i^2$$

where:

- a) $m = 33$ = number of variables;
- b) $l = 19$ = number of PCs we selected.

The pr.var we get in the previous part is the λ_i . Therefore, the g and h can be calculated. Thus, the control limit for SPE is 3.606688 and the control limit for T² is 30.14353

2. Use the fault detection control limits to perform fault detection on the cases with potential disturbances, plot the results and possibly calculate the false alarm rates and missed detection rates.

- 1) Monitor IDV(3):
 - a) Use R-studio to load the test data with process variables only, which is a 960×33 matrix.
 - b) Scale the test data with the mean and standard deviation of the train data.
 - c) Perform PCA to the scaled test data

```
> cumsum(pve)
```

```

[1] 0.2957711  0.4349313  0.4998394  0.5482258
[5] 0.5929878  0.6349203  0.6754617  0.7149859
[9] 0.7508743  0.7796961  0.8057290  0.8305107
[13] 0.8538739  0.8742898  0.8940247  0.9129949
[17] 0.9311234  0.9467499  0.9608350  0.9735690
[21] 0.9833132  0.9902037  0.9955991  0.9976906
[25] 0.9986893  0.9992971  0.9996249  0.9998557
[29] 0.9999342  0.9999964  1.0000000  1.0000000
[33] 1.0000000

```

Again, the number of PCs is selected as 19, which can explain more than 95% variance.

d) SPE statistic is calculated by $SPE = x^T (I - PP^T)x$.

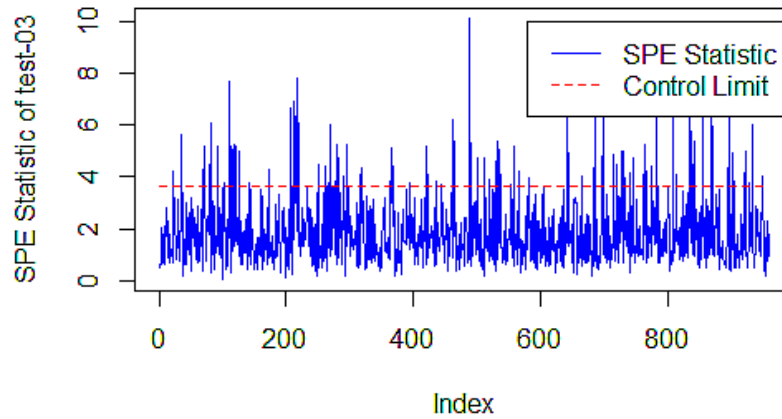


Fig. 28: SPE statistics of testing set of Process Data IDV(3)

```

> summary(Q<SPElimit)
  Mode  FALSE  TRUE  NA's
logical    89   871     0

```

From the value and plot above, we can see most samples are within the control limit of SPE.

e) T^2 statistic is calculated by $T^2 = x^T P \Lambda^{-1} P^T x$

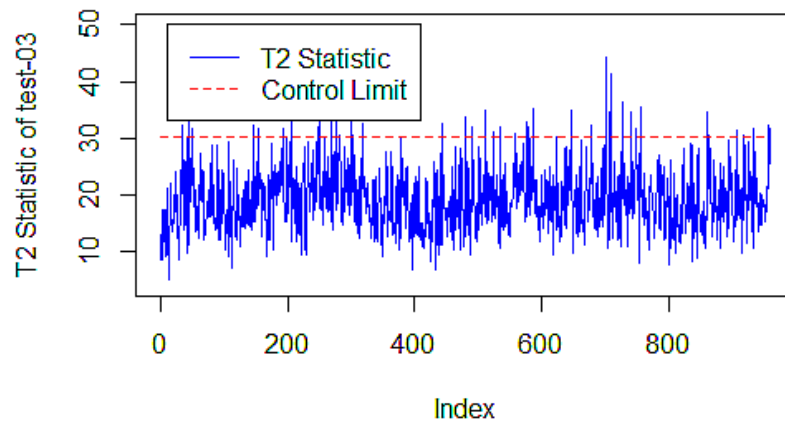


Fig. 29: T^2 statistics of testing set of Process Data IDV(3)

```
> summary(T2<T2limit)
```

Mode	FALSE	TRUE	NA's
logical	45	915	0

From the value and plot above, we can see most samples are within the control limit of T^2 .

2) Monitor IDV(5):

```
> cumsum(pve)
```

```
[1] 0.9155511 0.9976481 0.9987435 0.9993210
[5] 0.9995499 0.9996478 0.9996901 0.9997295
[9] 0.9997628 0.9997956 0.9998269 0.9998538
[13] 0.9998793 0.9999036 0.9999263 0.9999469
[17] 0.9999647 0.9999809 0.9999905 0.9999959
[21] 0.9999987 0.9999999 1.0000000 1.0000000
[25] 1.0000000 1.0000000 1.0000000 1.0000000
[29] 1.0000000 1.0000000 1.0000000 1.0000000
[33] 1.0000000
```

The number of PCs is selected as 2, which can explain more than 95% variance.

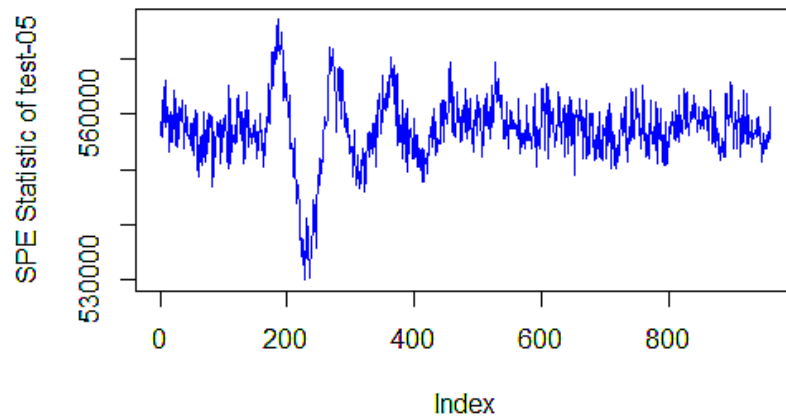


Fig. 30: SPE statistics of testing set of Process Data IDV(5)

```
> summary(Q<SPElimit)
  Mode  FALSE   TRUE  NA's
logical   960     0     0
```

From the value and plot above, we can see all samples are beyond the control limit of SPE.

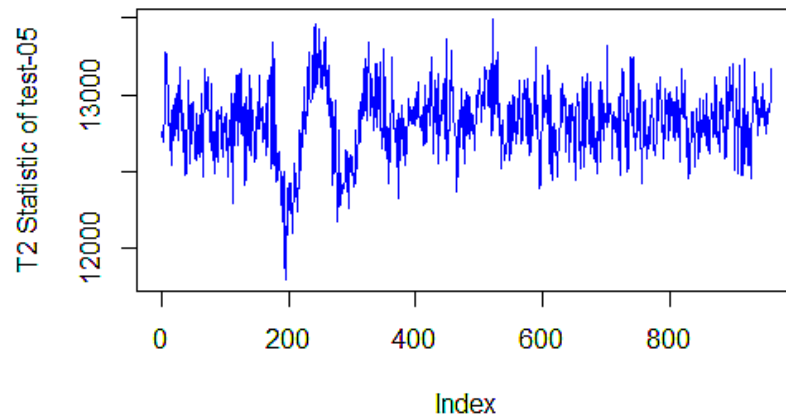


Fig. 31: T^2 statistics of testing set of Process Data IDV(5)

```
> summary(T2<T2limit)
  Mode  FALSE   TRUE  NA's
logical   960     0     0
```

From the value and plot above, we can see all samples are beyond the control limit of T^2 .

3) Monitor IDV(9):

```
> cumsum(pve)
[1] 0.2568305 0.4209292 0.4938959 0.5436936
[5] 0.5885496 0.6300190 0.6702474 0.7081278
[9] 0.7450807 0.7778206 0.8048244 0.8288412
[13] 0.8516668 0.8733200 0.8937784 0.9120195
[17] 0.9297450 0.9447609 0.9590127 0.9722949
[21] 0.9817732 0.9895353 0.9955881 0.9976882
[25] 0.9987013 0.9993317 0.9996359 0.9998414
[29] 0.9999293 0.9999972 1.0000000 1.0000000
[33] 1.0000000
```

The number of PCs is selected as 19, which can explain more than 95% variance.

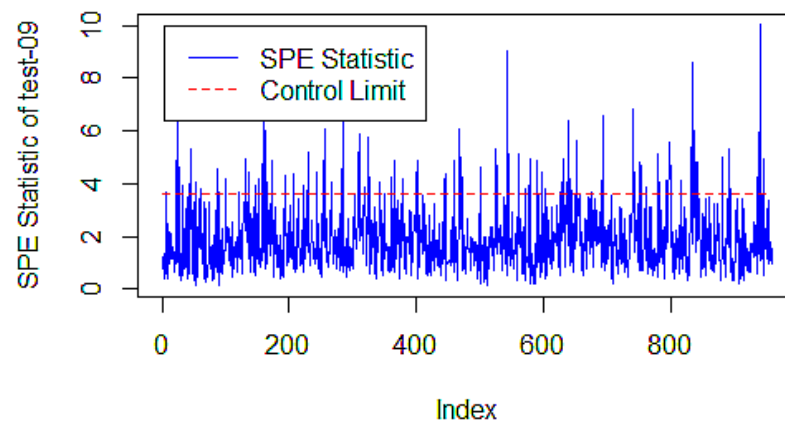


Fig. 32: SPE statistics of testing set of Process Data IDV(9)

```
> summary(Q<SPElimit)
  Mode  FALSE  TRUE  NA's
logical   100   860     0
```

From the value and plot above, we can see most samples are within the control limit of SPE.

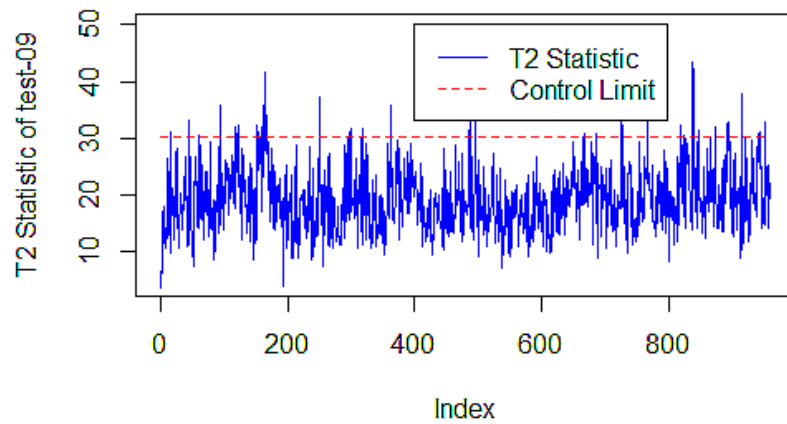


Fig. 33: T^2 statistics of testing set of Process Data IDV(9)

```
> summary(T2<T2limit)
```

Mode	FALSE	TRUE	NA's
logical	40	920	0

From the value and plot above, we can see most samples are within the control limit of T^2 .

4) Monitor IDV(13):

```
> cumsum(pve)
```

```
[1] 0.5811760 0.8639980 0.9565554 0.9723981
[5] 0.9796436 0.9837454 0.9865922 0.9881663
[9] 0.9895941 0.9908237 0.9919567 0.9930161
[13] 0.9939341 0.9947377 0.9954388 0.9960957
[17] 0.9966726 0.9972216 0.9977394 0.9981913
[21] 0.9986254 0.9990072 0.9993508 0.9995971
[25] 0.9998056 0.9999512 0.9999862 0.9999940
[29] 0.9999963 0.9999983 1.0000000 1.0000000
[33] 1.0000000
```

The number of PCs is selected as 3, which can explain more than 95% variance.

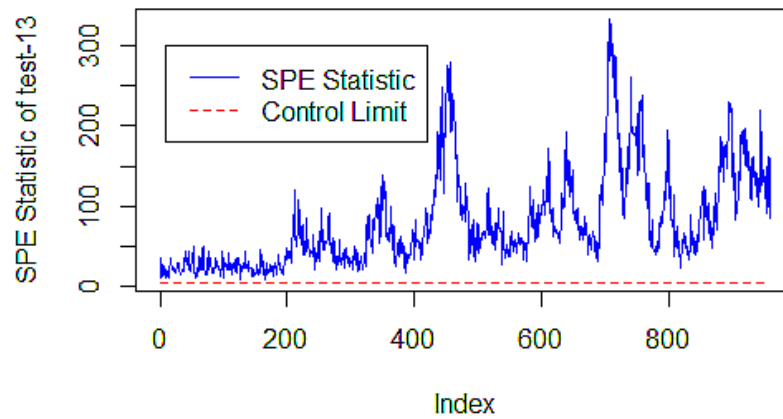


Fig. 34: SPE statistics of testing set of Process Data IDV(13)

```
> summary(Q<SPElimit)
  Mode  FALSE  TRUE  NA's
logical   960    0     0
```

From the value and plot above, we can see all samples are beyond the control limit of SPE.

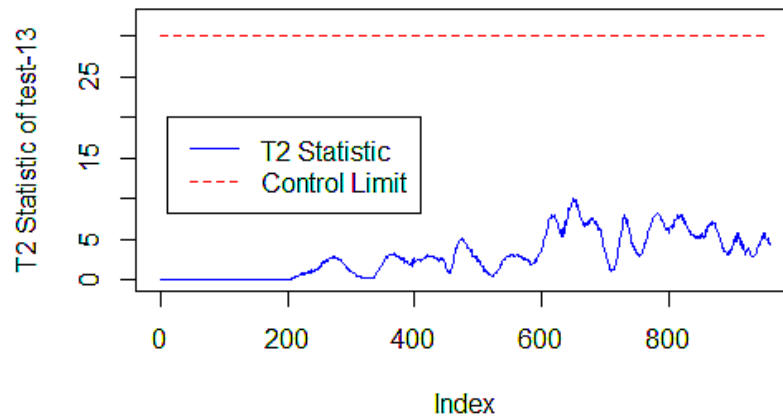


Fig. 35: T^2 statistics of testing set of Process Data IDV(13)

```
> summary(T2<T2limit)
  Mode  FALSE  TRUE  NA's
logical    0   960     0
```

From the value and plot above, we can all samples are within the control limit of T^2 .

5) Monitor IDV(14):

```
> cumsum(pve)
[1] 0.6932790 0.9313744 0.9441352 0.9520595
[5] 0.9567311 0.9612488 0.9654586 0.9693010
[9] 0.9728744 0.9760337 0.9790396 0.9818937
[13] 0.9843684 0.9867117 0.9887456 0.9906793
[17] 0.9925218 0.9942231 0.9957396 0.9971655
[21] 0.9982181 0.9989452 0.9995927 0.9997791
[25] 0.9998797 0.9999351 0.9999640 0.9999849
[29] 0.9999933 0.9999999 1.0000000 1.0000000
[33] 1.0000000
```

The number of PCs is selected as 4, which can explain more than 95% variance.

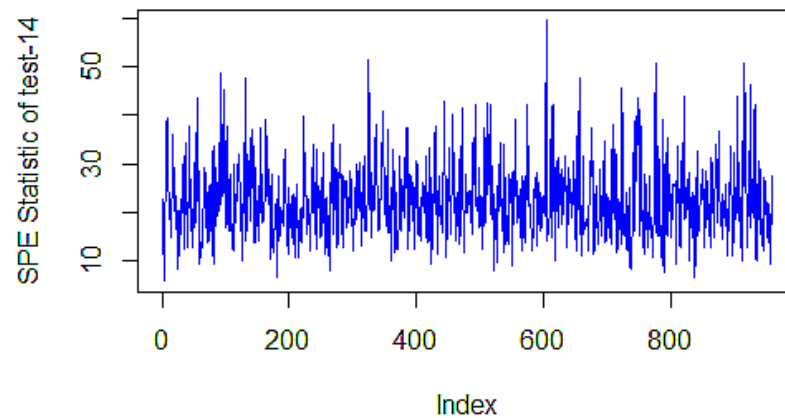


Fig. 36: SPE statistics of testing set of Process Data IDV(14)

```
> summary(Q<SPElimit)
  Mode FALSE  TRUE  NA's
logical  960    0     0
```

From the value and plot above, we can see all samples are beyond the control limit of SPE.

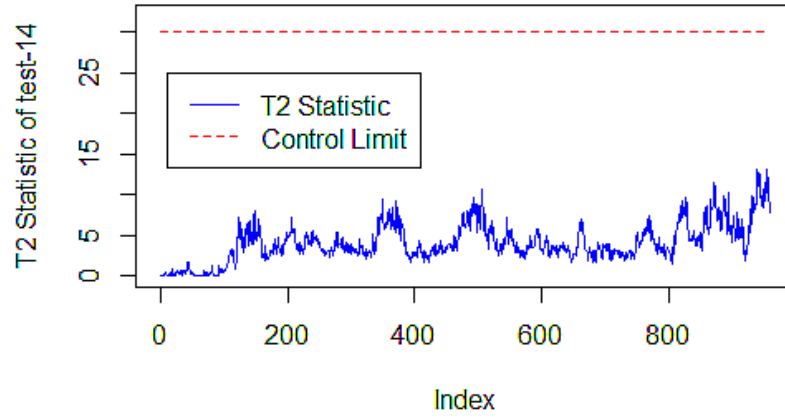


Fig. 37: T^2 statistics of testing set of Process Data IDV(14)

```
> summary(T2<T2limit)
```

```
Mode   FALSE   TRUE   NA's
logical 0     960    0
```

From the value and plot above, we can see all samples are within the control limit of T^2

We make a summary by using the following tables:

Results of quality monitoring	% of Samples Beyond SPElimit	% of Samples Beyond T2limit	Classification
IDV (3)	4.69	7.81	Quality-Irrelevant
IDV (5)	100	100	Quality-Relevant
IDV (9)	5.21	6.25	Quality-Irrelevant
IDV (13)	71.88	0	Quality-relevant
IDV (14)	6.77	3.13	Quality-Irrelevant

Table 2

Results of process monitoring	% of Samples Beyond SPElimit	% of Samples Beyond T2limit
IDV (3)	9.27	4.69
IDV (5)	100	100
IDV (9)	10.41	4.17
IDV (13)	100	0
IDV (14)	100	0

Table 3

Also we calculate the false alarm rates(FAR) and missed detection rate(MDR) by the following formula s:

$$FAR = \frac{\text{Number of samples } (T^2 \text{ index} > \text{control limit, fault} = 0)}{\text{Number of samples (fault} = 0)}$$

$$\text{MDR} = \frac{\text{Number of samples (T}^2 \text{ index} < \text{control limit, fault} \neq 0)}{\text{Number of samples (fault} \neq 0)}$$

Where:

(fault = 0): # of samples that are not fault in the quality monitoring;

(fault $\neq$ 0): # of samples that are fault in the quality monitoring;

(T² index > control limit): # of samples whose T² exceeds the control limit in process monitoring;

(T² index < control limit): # of samples with T² within the control limit in process monitoring.

FAR and MDR for IDV(3) can be calculated while the similar approach can be applied to obtain them for the other four disturbed datasets, namely IDV(5), IDV(9), IDV(13), and IDV(14).

```
> dim( d03teOriginal_1)
```

```
[1] 44 40
```

Thus the # of samples that are not fault in the quality monitoring but beyond the control limit in process monitoring for IDV(3) is 44.

Therefore, the FAR= 44 / 885 = 4.97%

```
> dim( d03teOriginal_1)
```

```
[1] 74 40
```

Thus the # of samples that are fault in the quality monitoring but within the control limit in process monitoring for IDV(3) is 74.

Therefore, the MDR= 74 / 75 = 98.67%

Finally, we make a summary as below:

Rates	IDV(3)	IDV(5)	IDV(9)	IDV(13)	IDV(14)
FAR	44/885 = 4.97%		38/900 = 4.22%	0/960 = 0	0/930 = 0
MDR	74/75 = 98.67%	0/960 = 0	58/60 = 96.67%		30/30 = 100%

Table 4

- FAR can be considered as the ratio between the number of negative events wrongly categorized as positive (false positives) and the total number of positive events. On the other hand, MDR can be considered as the ratio between the number of positive events wrongly categorized as negative (false negatives) and the total number of negative events;
- Large MDR shows that for the processes that lead to bad results, most of them are actually correct, which indicates that there is no direct relationship between process and result. Therefore, from the above table, we can predict that IDV(3), IDV(9) and IDV(14) are irrelevant to the quality;
- As for IDV(5) and IDV(13), zero MDR and zero FAR indicate that all wrong process will bring us bad results, which make IDV(5) and IDV(13) definitely relevant to the quality.

The above conclusions agree with that we obtain in task 1 and Fig.2.

3. Use the LDA classification method between the normal case and some of the cases with disturbances and see how overlapped they are, and plot the results.

Code Sample:

```
LDAX = LDA_IDVx_2
```

```
ldax = LDAX[,1:34]
```

```
library(MASS)
```

```
lda.fit = lda(Label ~., data = ldax)
```

```
plda = predict(object=lda.fit,newdata=ldax)
```

```
plot(plda$x,col=plda$class)
```

```
plot(plda$x,col=ldax$Label+1)
```

1) IDV(3):

Result:

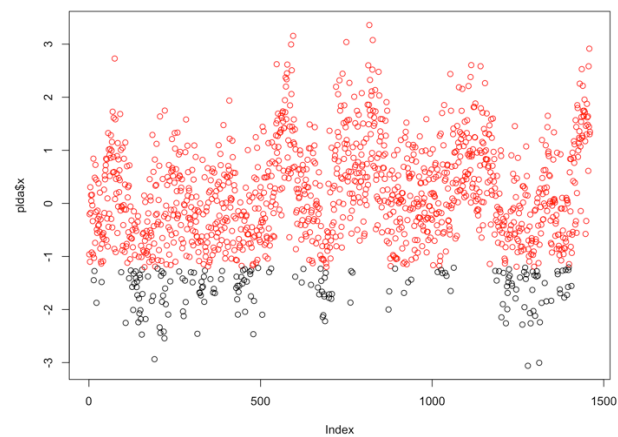


Fig. 38: LDA result for IDV(3)

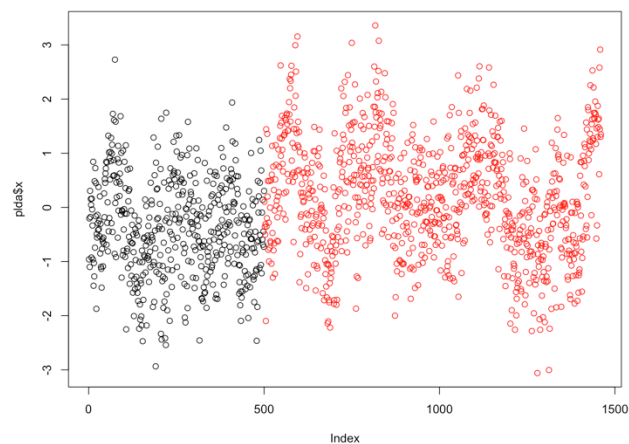


Fig. 39: Actual result for IDV(3)

From the plotting above, we can conclude that the data from IDV3 are highly overlapped with the normal data. As a result, we can not separate these two groups of data by LDA.

2) IDV(5):

Result:

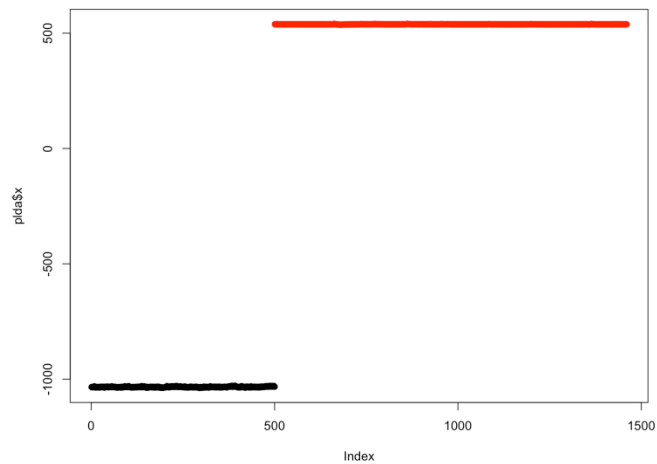


Fig. 40: LDA result for IDV(5)

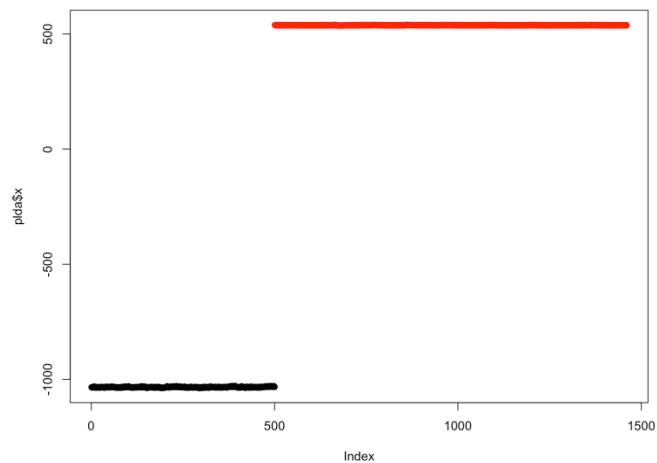


Fig. 41: Actual result for IDV(5)

From the plotting above, we can conclude that the data from IDV5 are not overlapped with the normal data. As a result, we can separate these two groups of data by LDA.

3) IDV(9):

Result:

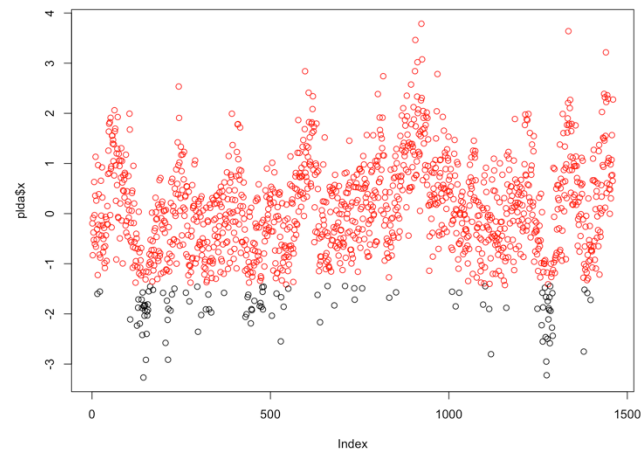


Fig. 42: LDA result for IDV(9)

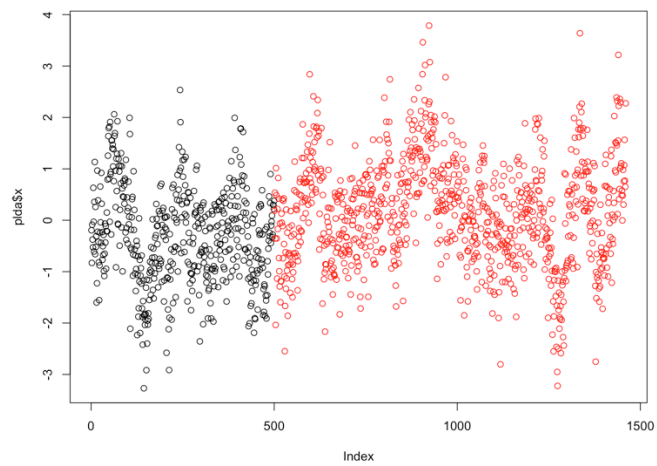


Fig. 43: Actual result for IDV(9)

From the plotting above, we can conclude that the data from IDV9 are highly overlapped with the normal data. As a result, we can not separate these two groups of data by LDA.

4) IDV(13):

Result:

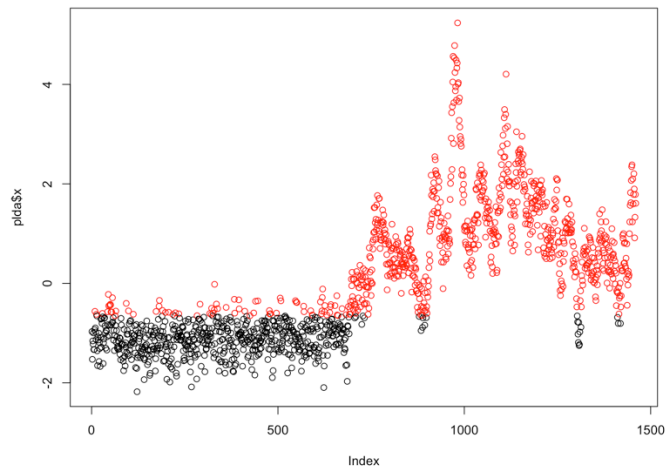


Fig. 44: LDA result for IDV(13)

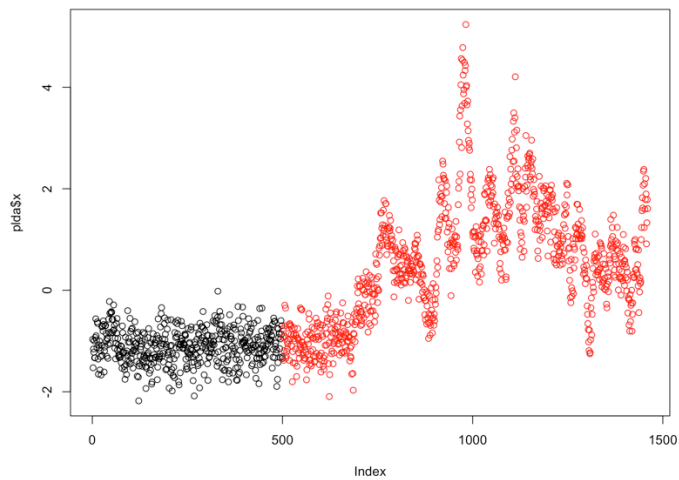


Fig. 45: Actual result for IDV(13)

From the plotting above, we can conclude that the data from IDV13 are overlapped with the normal data. We can not separate these two groups of data by LDA, but the result is better than the other quality-irrelevant groups.

5) IDV(14):

Result:

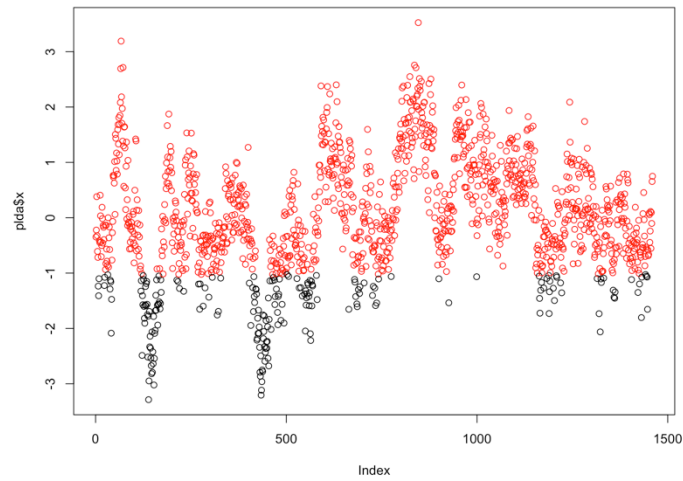


Fig. 46: LDA result for IDV(14)

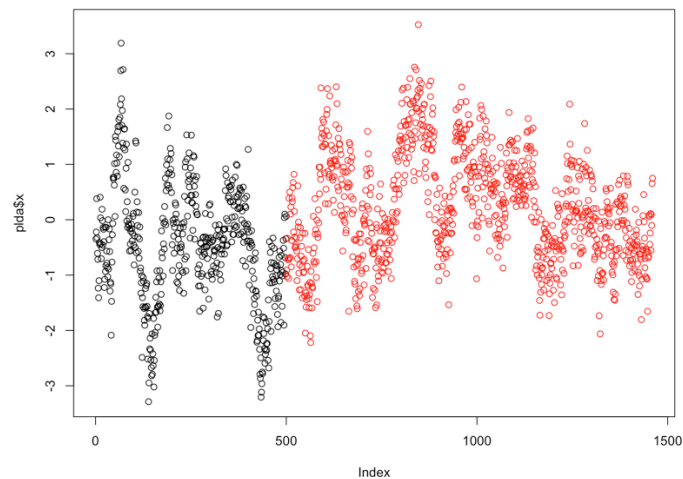


Fig. 47: Actual result for IDV(14)

From the plotting above, we can conclude that the data from IDV14 are highly overlapped with the normal data. As a result, we can not separate these two groups of data by LDA.

- 6) Mixed:
Result:

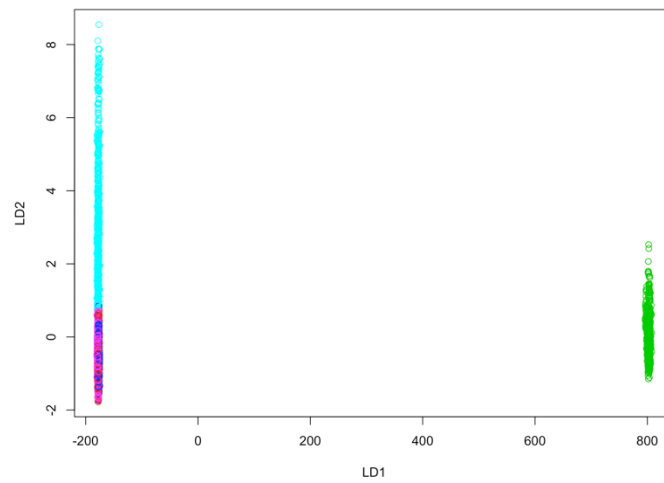


Fig. 48: LDA result for all groups(LD1&LD2)

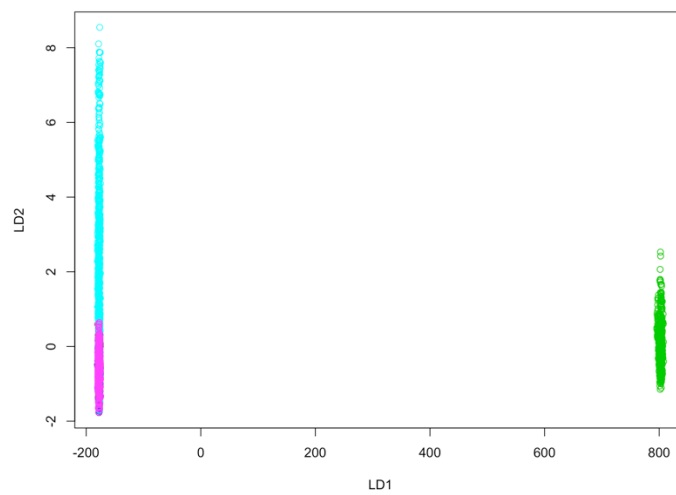


Fig. 49: Actual result for all groups(LD1&LD2)

The first two LDs can describe 99% variance.

We can also use LD2 & LD3 to plot.

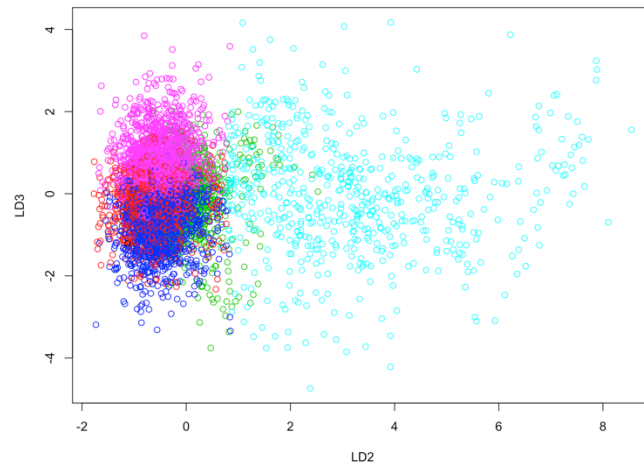


Fig. 50: LDA result for all groups(LD2&LD3)

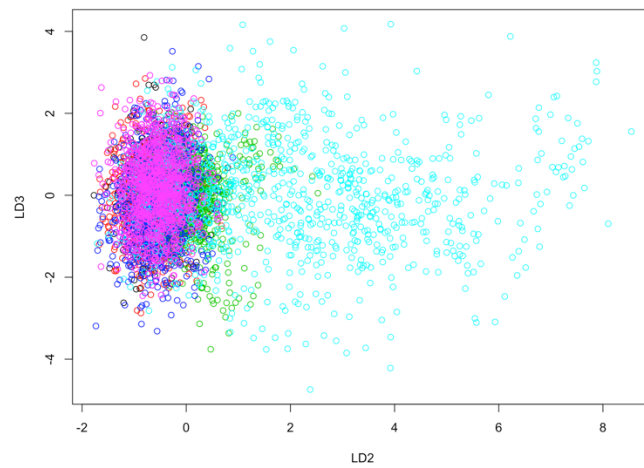


Fig. 51: Actual result for all groups(LD2&LD3)

From the plotting above, we can conclude that only one group of data (IDV5, green) are separated by LDA(LD1&LD2). And also, most of data from IDV13(baby blue) can be separated by LDA(LD2&LD3).

Task 3 – CCA Using Quality Data (XMEAS(37-41))

1. Perform CCA using the code from the TA on the normal case data; use regularization if necessary, and use cross-validation to determine the amount of regularization.

After cross-validation, CCA factors=1

2. Plot the Y predictions vs. actual data, and plot the residuals. Calculate the R^2 values.

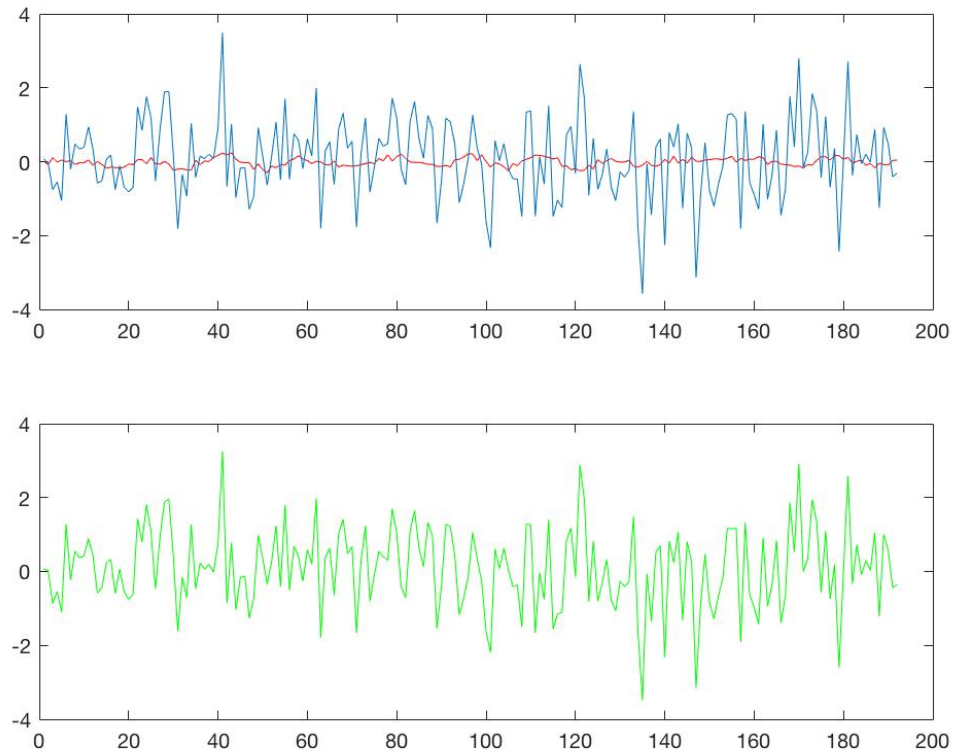


Fig. 52: CCA based combined plot for IDV(3)

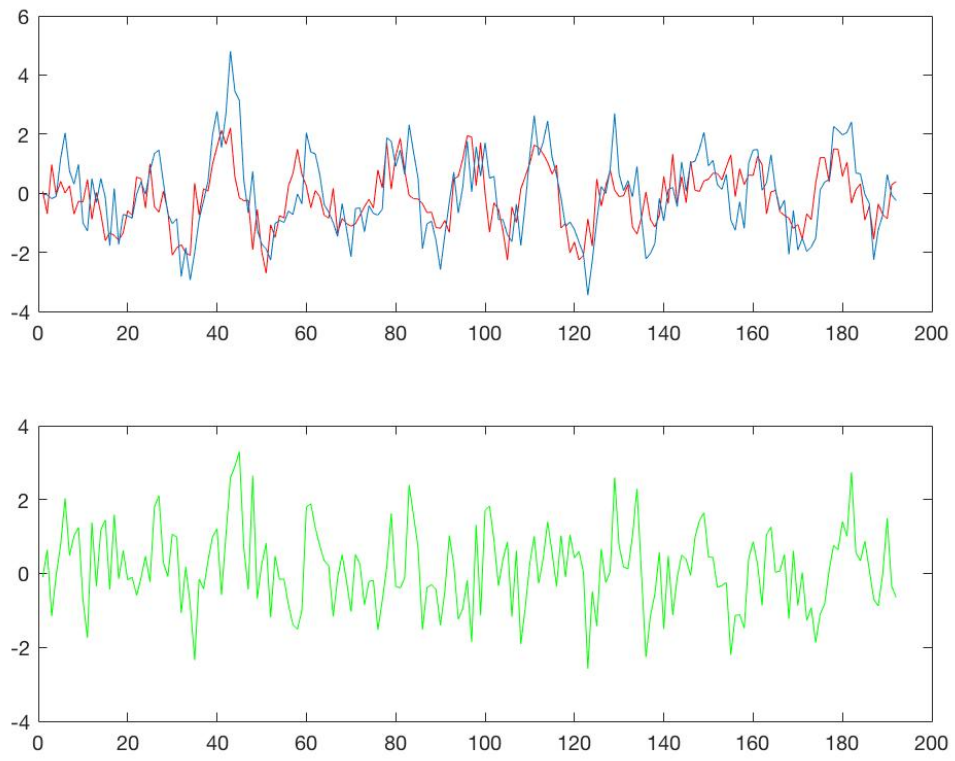


Fig. 53: CCA based combined plot for IDV(5)

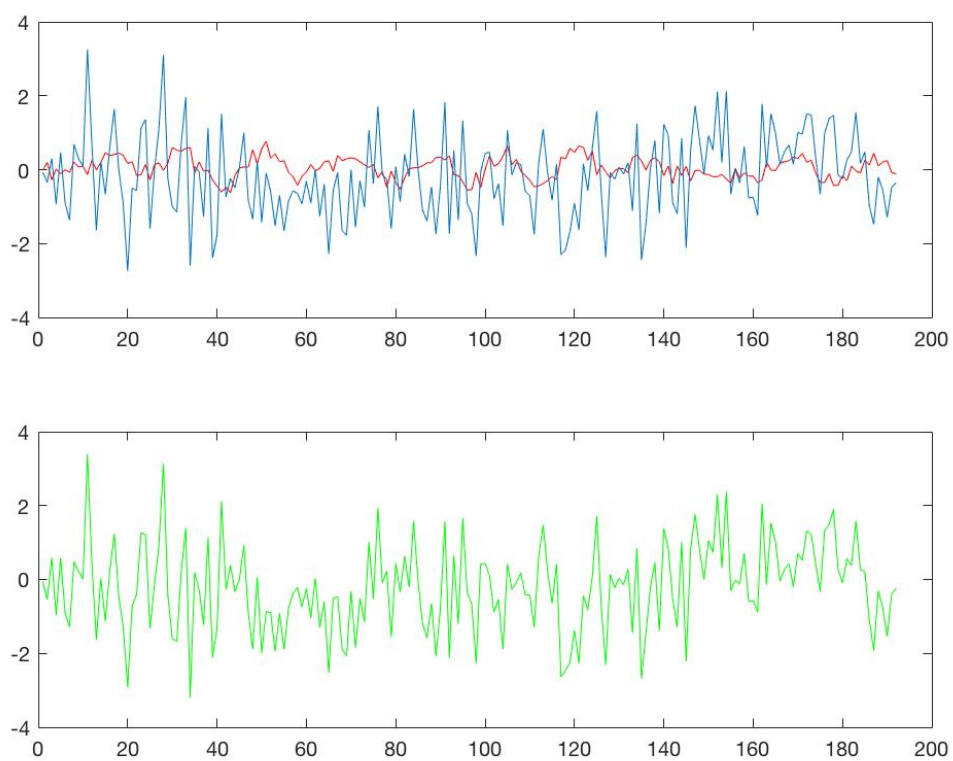


Fig. 54: CCA based combined plot for IDV(9)

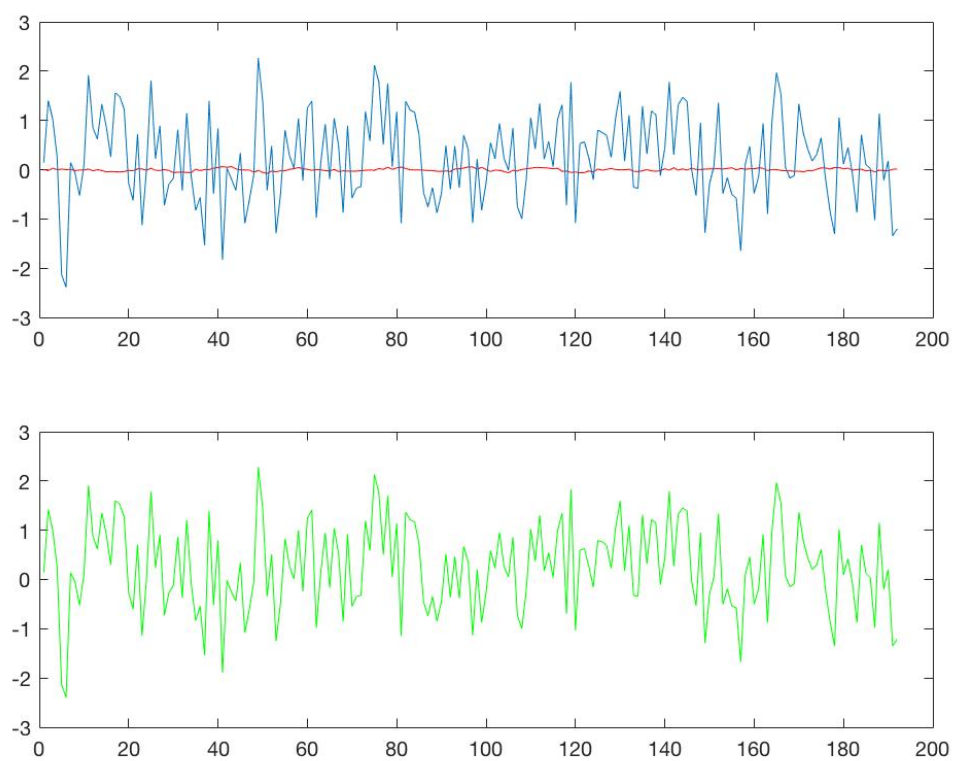


Fig. 55: CCA based combined plot for IDV(13)

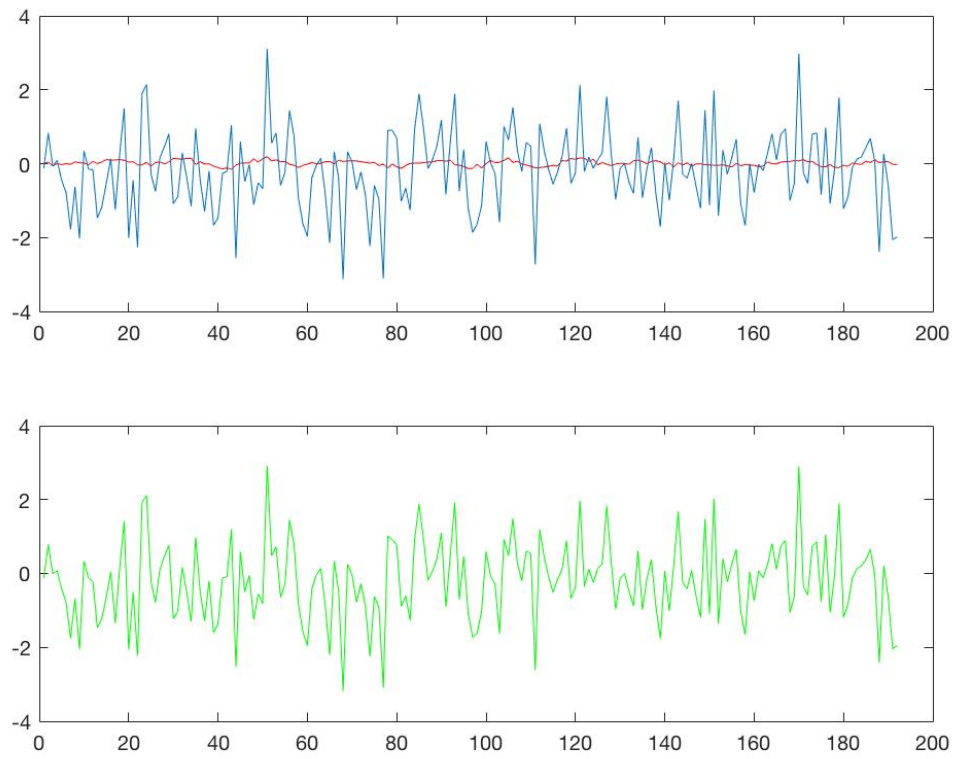


Fig. 56: CCA based combined plot for IDV(14)

And $R^2 = 0.0563$

3. Calculate the X residuals after the CCA factors are extracted. These are the basis for performing Concurrent CCA monitoring.

Task 4 – Concurrent CCA Monitoring Using Quality Data (XMEAS(37-41))

1. Perform CCCA monitoring using the CCCA code from the TA on the cases with disturbances. This includes subsequent PCA on the X-residuals and Y-residuals.
2. Establish control limits for the co-variation (T_c^2), the X-residual PCA, and the Y-residual PCA. Plot the control charts.

IDV(3)

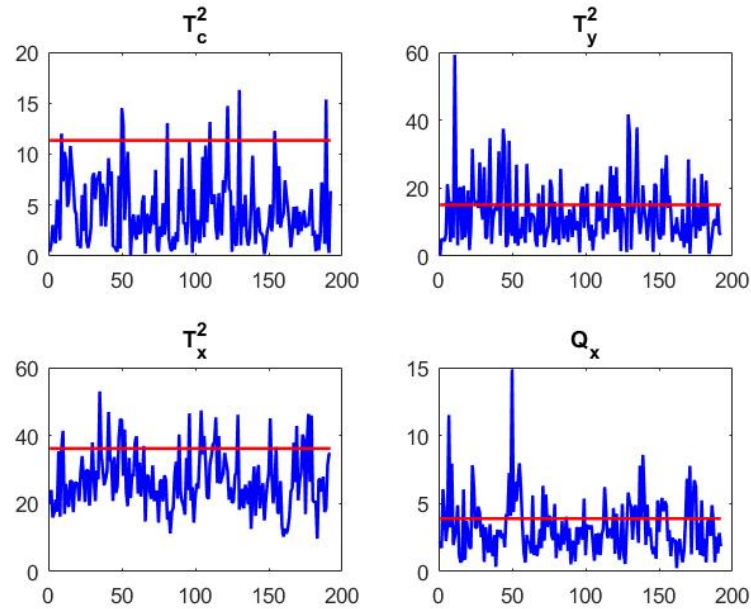


Fig. 57

The CCCA monitoring result indicates that T_c^2 , T_x^2 , and T_y^2 have more data below the control limits. Therefore, the fault is quality irrelevant, which should receive less attention in the process.

IDV(5)

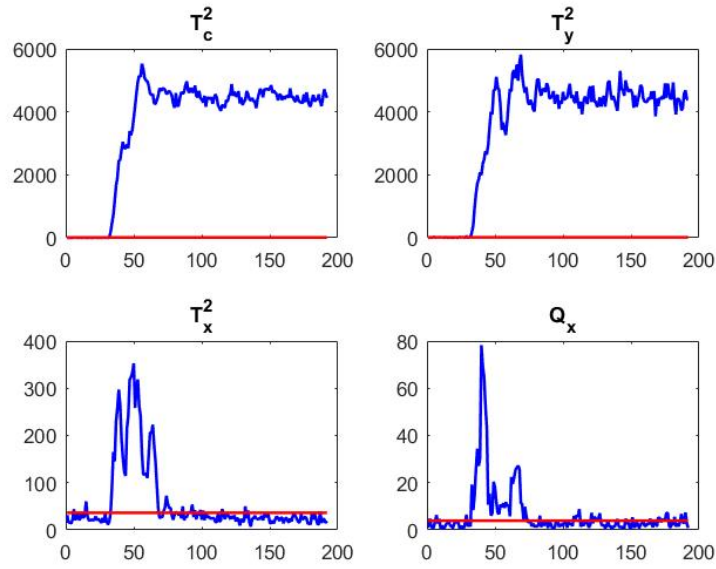


Fig. 58

The CCCA monitoring result indicates that T_{c2} , T_{x2} , and T_{y2} have more data above the control limits. Therefore, the fault is quality relevant, which should receive less attention in the process.

IDV(9)

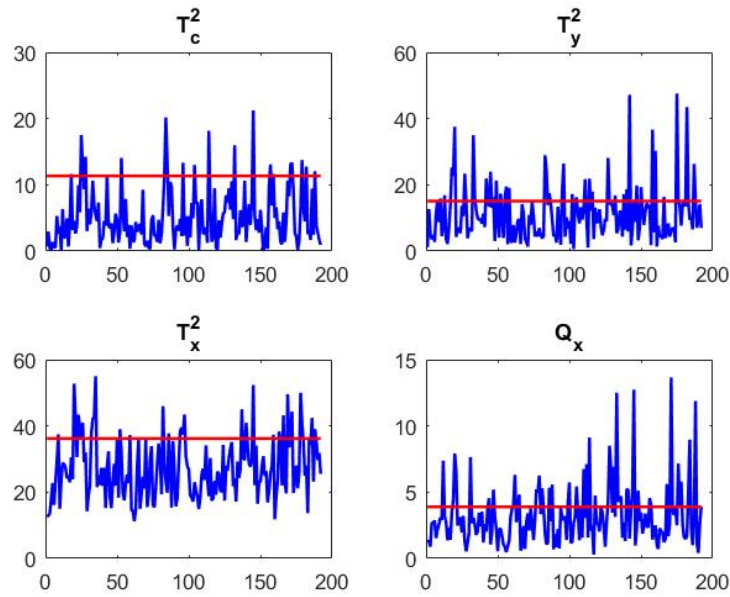


Fig.59

The CCCA monitoring result indicates that T_c^2 , T_x^2 , and T_y^2 have more data below the control limits. Therefore, the fault is quality irrelevant, which should receive less attention in the process.

IDV(13)

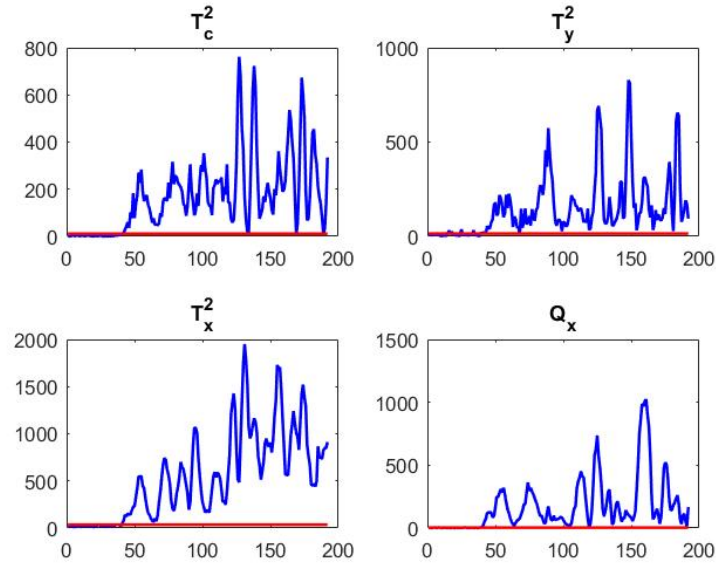


Fig. 60

The CCCA monitoring result indicates that T_c^2 , T_x^2 , and T_y^2 have more data above the control limits. Therefore, the fault is quality relevant, which should receive less attention in the process.

IDV(14)

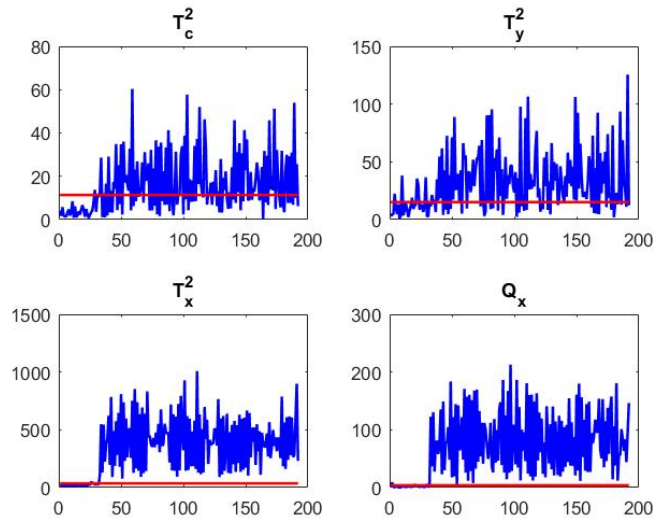


Fig. 61

The CCCA monitoring result indicates that T_c^2 , T_x^2 , and T_y^2 have more data above the control limits. Therefore, the fault is quality relevant, which should receive less attention in the process.

iii. Conclusion & Discussion

The differences between PCA/LDA

- 1) PCA finds the dimension in which the reflection of data have the biggest variance. But LDA finds the dimension in which the data can be separated (like Fig.1).

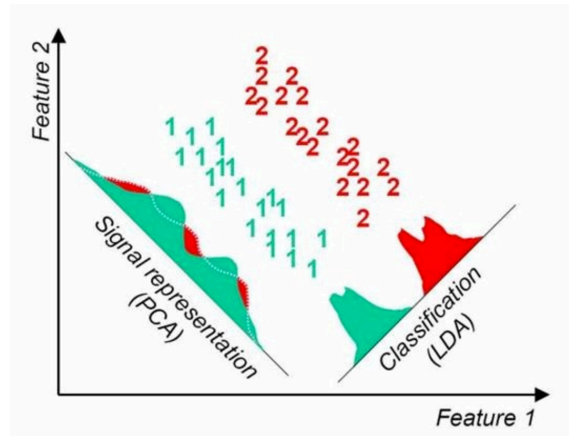


Fig. 62

- 2) If two groups of data are highly overlapped, which means, the mean of data is almost equal and the variance is very large compare to the mean, LDA cannot separate these two groups of data (like Fig.2). Thus, in my opinion, this is the reason why we cannot separate disturbance data from the normal data.

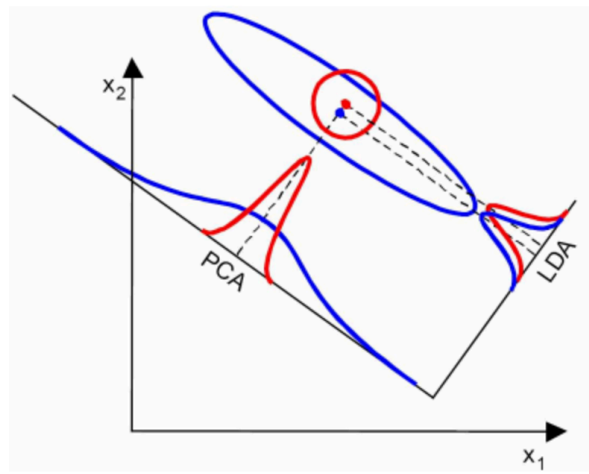


Fig. 63

Reference

- [1] Q. Zhu, Q. Liu, and S. J. Qin (2016). Concurrent canonical correlation analysis modeling for quality-relevant monitoring, Proc. Of IFAC Symposium on DYCOPS-CAB, IFAC-Papers-OnLine, vol. 49, no. 7, pp. 1044–1049
- [2] J. J. Downs and E. F. VOGEL (1993). A plant-wide industrial process control problem. Computers chem. Engng. Vol. 17, No. 3. Pp.245-255
- [3] James, G., Witten, D., Hastie, T., Tibshirani, R. (2013), An Introduction to Statistical Learning with Applications in R.
- [4] S. J. Qin(2017), CHE 599 course handouts